5 **(a)** State the principle of superposition.

.....

.....

..... [2]

(b) An interference pattern is produced with the arrangement shown in Figure 5.1.

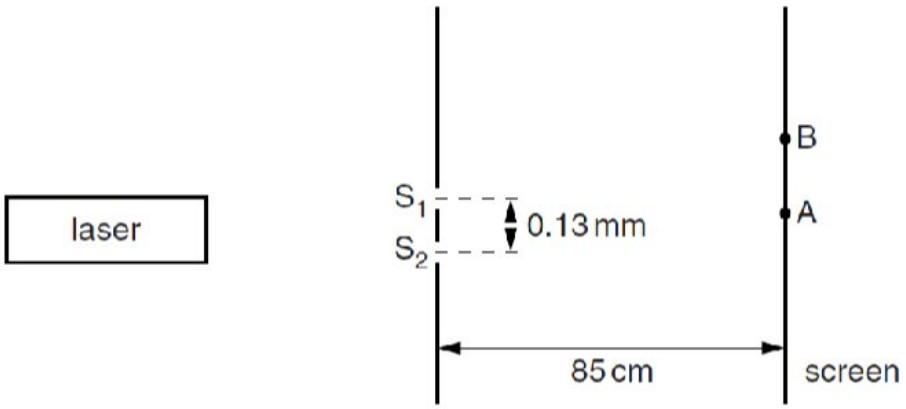


Figure 5.1 (not to scale)

Laser light of wavelength λ of 546 nm is incident on the slits S_1 and S_2. The slits are a distance 0.13 mm apart. The distance between the slits and the screen is 85 cm.

Two points on the screen are labelled A and B. The path difference between S_1A and S_2A is zero. The path difference between S_1B and S_2B is 2.5 λ. Maxima and minima of intensity of light are produced on the screen.

(i) Explain why a maxima is produced at point A.

.....

.....

.....

..... [2]

(ii) Calculate the distance AB.

distance = ... m [3]

(iii) The laser is replaced by a laser emitting blue light. State and explain the change in the distance between the maxima observed on the screen.

.....

..... [1]

(iv) The intensity of the laser passing through the two slits was initially the same. The intensity of the light through **one** of the slits is now reduced. Describe the changes, in any, in the appearance of the fringes on the screen.

.....

.....

.....

..... [2]

[Total: 10]